

COURSE TITLE : OPTICAL INSTRUMENTATION
COURSE CODE : 5214
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER: 52
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
1	Light and its Properties	14
2	Fibre Optic Instrumentation	14
3	Laser Fundamentals	13
4	Laser Applications	11
TOTAL		52

COURSE OUTCOME

Module	G.O	On completion of the study of this Module the student will be able
1	1	To Understand the phenomena of Reflection, Refraction, and Dispersion.
	2	To Understand the phenomena of Interference and Diffraction
	3	To Understand phenomena of Polarization.
2	1	To Basics of Fibre optic Communication.
	2	To Understand fibre optic sensors
	3	To Understand optoelectronic Components
3	1	To Understand the Laser Fundamentals.
	2	To Understand the construction and working of different types of Lasers.
4	1	To Understand Important Applications of Laser
	2	To Understand Application of Laser for Measurement

On the completion of the study the student will be able

Specific Outcome

MODULE I LIGHT AND ITS PROPERTIES

1.1.0 To understand the phenomena of Reflection, Refraction, and Dispersion.

- 1.1.1 To explain the phenomenon and laws of reflection
- 1.1.2 To describe the phenomenon and laws of Refraction.
- 1.1.3 To explain the phenomenon of Dispersion.

1.2.0 To Understand the phenomena of Interference and Diffraction

- 1.2.1 To explain the phenomenon of interference.
- 1.2.2 To describe the concept of Phase difference and path difference.

- 1.2.3 To explain the condition for bright and dark fringes in interference.
- 1.2.4 To explain the distribution of intensity in interference pattern.
- 1.2.5 To describe the formation of Newton's Rings.
- 1.2.6 To explain the use of Newton's ring to measure the radius of curvature of the lens.
- 1.2.7 To describe the phenomenon of Diffraction.
- 1.2.8 To explain the difference between Interference and Diffraction.

1.3.0 To Understand the phenomena of Polarization

- 1.3.1 To define the phenomenon of Polarization
- 1.3.2 To explain the laws of Polarization.
- 1.3.3 To explain the application of Polarization in Analytical Instrumentation.

MODULE II FIBER OPTIC INSTRUMENTATION

2.1.0 To Understand Fibre Optic Communication.

- 2.1.1 To describe the advantages of fibre optic communication.
- 2.1.2 To Explain the Principle of Optical Fibres.
- 2.1.3 To describe optical fibre construction.
- 2.1.4 To explain acceptance angle and Numerical aperture.
- 2.1.5 To describe the modes of operation of optical fibres.
- 2.1.6 To describe Single mode step index fibre, multi mode step index fibre and multimode graded index fibre and their comparison.
- 2.1.7 To explain with a block diagram, Fibre optic Communication System.

2.2.0 To understand fibre optic Sensors.

- 2.2.1 To describe the working of fibre optic displacement sensor.
- 2.2.2 To explain the working of fibre optic Pressure sensor.
- 2.2.3 To describe the working of fibre optic Temperature sensor.
- 2.2.4 To explain the working of fibre optic Force sensor.
- 2.2.5 To describe the working of fibre optic Torque sensor.
- 2.2.6 To explain the working of fibre optic Strain sensor.
- 2.2.7 To describe the working of fibre optic Viscosity sensor.
- 2.2.8 To explain the working of fibre optic Fluid Level sensor.
- 2.2.9 To describe the working of fibre optic Fluid Flow sensor.

2.3.0 To understand optoelectronic components.

- 2.3.1 To describe the working of photo diode.
- 2.3.2 To describe the working of Photo Transistor.
- 2.3.3 To explain the construction and operation of PiN Diode.
- 2.3.4 To explain the construction and operation of Avalanche Photo Diode.
- 2.3.5 To describe the principle of operation of LED.
- 2.3.6 To describe the working of LED drive circuit.

MODULE III LASER FUNDAMENTALS

3.1.0 To Understand Lasers

- 3.1.1 To describe the characteristics of LASER beam.
- 3.1.2 To describe absorption, spontaneous emission and stimulated emission.

- 3.1.3 To describe the population Inversion.
- 3.1.4 To explain the basic requirements of producing LASER.
- 3.1.5 To describe the classification of LASER.

3.2.0 Construction and Working of Different types of lasers

- 3.2.1 To describe the construction and operation of Ruby Laser.
- 3.2.2 To explain the construction and operation of Nd-YAG Laser.
- 3.2.3 To describe the construction and operation of He-Ne Laser.
- 3.2.4 To explain the construction and operation of Co2 Laser.
- 3.2.5 To describe the construction and operation of Argon Laser.
- 3.2.6 To explain the construction and operation of Liquid Dye Laser.
- 3.2.7 To describe the construction and operation of Semiconductor Laser.

MODULE IV LASER APPLICATIONS

4.1.0 To Understand Important Laser Applications.

- 4.1.1 To describe the Industrial Applications of Laser.
- 4.1.2 To explain the Medical Applications of Laser.
- 4.1.3 To describe the Scientific applications of Laser
- 4.1.4 To explain the Military applications.
- 4.1.5 To illustrate the Miscellaneous Applications

4.2.0 To understand Laser application in Instrumentation.

- 4.2.1 To explain laser based displacement measuring.
- 4.2.2 To explain laser Doppler velocity meter.

CONTENT DETAILS

MODULE I

Phenomenon and laws of reflection - refraction - total internal reflection – critical angle, - dispersion - interference- Young's double slit experiment- phase difference and path difference-constructive and destructive interference- conditions for bright and dark fringes- conditions for distribution of light intensities – concept of coherence- Newton's Rings - Application of Newton's rings - diffraction – difference between interference and diffraction- polarization- polarizer and analyzer- Laws of polarization- Malus law and Brewsters law.

MODULE II

Advantages of fibre optic communication- Principle of Optical Fibres- optical fibre construction- acceptance angle and Numerical aperture- modes of operation of optical fibres- Single mode step index fibre, multi mode step index fibre and multimode graded index fibre and - comparison- block diagram- Fibre optic Communication System - passive and active optical fibre sensors-displacement sensor- Pressure sensor- Temperature sensor- Force sensor- Torque sensor- Strain sensor- Viscosity sensor- Level sensor- Flow sensor. Construction and working of photo diode- PiN diode- Avalanche photo diode- photo transistor- LED principle of operation, LED drive circuits.

MODULE III

Characteristics of laser beam- Interaction between matter and radiation – absorption- spontaneous emission- stimulated emission – Thermal equilibrium- Population inversion- Basic requirements for producing laser- Classification of laser - Construction and Working of - Ruby laser- Nd-YAG laser- He-Ne Laser- Argon Ion Laser-CO₂ Laser- Liquid dye Laser- Semiconductor Laser.

MODULE IV

Industrial Applications –laser welding- cutting and laser drilling- Medical applications of laser- Scientific applications of laser- Military applications of laser- Miscellaneous applications of laser- (laser printer- optical disc recording and reading- holography – construction of hologram and reconstruction of image)- Laser application in Instrumentation-displacement measurement - Laser Doppler velocity meter

REFERENCES

1. S.Nagabhushana & N. Sathyanarayana - Lasers and Optical Instrumentation – I K International Publishing House
2. N. Subrahmanyam & Brijlal - A Text Book of Optics – S Chand
3. B.L.Theraja - Engineering Physics – S Chand
4. Keiser- Optical Fiber Communication System - TMH
5. K.Thyagaraja & Ghatak - Laser theory & application – Springer
6. M.N.Auadhnulu - An introduction to Lasers – S Chand